ModelIt!

Teacher's Guide

Why Do Some Animals Look Different?

Inherited Traits vs. Learned Behaviors

Quick Reference

Grade Level	Grade 3
NGSS Standard	3-LS3-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 3-LS3-1

Analyze and interpret data to provide evidence that plants and animals have traits inherited from parents and that variation of these traits exists in a group of similar organisms.

Three-Dimensional Learning

Science Practice	**Analyzing and Interpreting Data** — Students analyze data about parent and offspring traits to identify patterns of inheritance and explain why variation exists within groups of similar organisms.
Disciplinary Core Idea	**LS3.A Inheritance of Traits** — Many characteristics of organisms are inherited from their parents. Other characteristics result from individuals' interactions with the environment, which can range from diet to learning.
Crosscutting Concept	**Patterns** — Students identify patterns in how traits are passed from parents to offspring and how variation creates predictable differences within groups.

CAST Testing Alignment

MULTIPLE CHOICE:

Two brown dogs have a litter of puppies. Most puppies are brown, but one is golden. Which best explains why one puppy looks different?

CONSTRUCTED RESPONSE:

A student has brown eyes like her dad and curly hair like her mom, but her brother has blue eyes and straight hair. Using what you know about inherited traits, explain how two children from the same parents can look different. Use the words 'inherited' and 'variation' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Every living thing on Earth has traits — characteristics that make it unique. Understanding where traits come from helps us explain the incredible diversity of life, from the smallest flower to the largest whale.

What Are Traits?

Traits are observable characteristics of living things. In animals, traits include fur color, eye color, body size, ear shape, and behavior patterns. In plants, traits include flower color, leaf shape, and height. Some traits are structural (what the organism looks like), while others are behavioral (what the organism does). Every individual organism has its own unique combination of traits.

Inherited vs. Environmental Traits

Inherited traits are passed from parents to offspring through genes — microscopic instructions inside every cell. Eye color, hair texture, skin tone, and blood type are all inherited. Environmental traits are influenced by the world around the organism. A plant grown in shade will be taller and thinner than the same plant grown in sunlight. A dog that exercises a lot will be more muscular. These are changes caused by the environment, not by genes.

Why Siblings Look Different

Even though siblings share the same parents, each one inherits a different combination of genes. Think of it like mixing paint — the same two colors can be blended in different amounts to create many shades. Each child gets roughly half their genes from mom and half from dad, but WHICH half is random each time. This genetic shuffling creates variation. Identical twins are the exception — they share the exact same genes because they developed from one fertilized egg that split in two.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify parent traits as an external component and genetic variation and offspring appearance as internal components that change based on what traits the parents have.

E **Establish Relationships**
Students determine that parent traits drive genetic variation, and genetic variation determines offspring appearance — different parent combinations produce different levels of variation.

V **Visualize & Model**
Students build a model showing how parent traits combine through genetic variation to produce unique offspring appearances.

E **Evaluate Outcomes**
Students run similar parents, different parents, and mixed parents scenarios to observe how the diversity of parent traits affects variation and offspring appearance.

R **Revise & Extend**
Students add environmental factors to explore how things like food, sunlight, and temperature can change how inherited traits are expressed.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Look at these puppies — they are all from the same mom and dad, but they look totally different! How is that possible?"*

Watch For

Show photos of animal families with visible variation. Ask students to spot differences between siblings.

Slide 2: What Will We Learn Today?

3 min

Say: *"Today we are going to figure out why YOU look like your parents but not exactly like your brothers or sisters!"*

Watch For

Read objectives together. Quick activity: have students identify one trait from their mom and one from their dad.

Slide 3: The Big Question

3 min

Say: *"Why do puppies from the same litter look different? If they have the same parents, should not they all look the same?"*

Watch For

Turn and talk: Do you look exactly like your siblings? Why or why not? Share answers.

Slide 4: Let's Build a Model!

3 min

Say: *"We are going to build a model to crack this mystery of why siblings look different!"*

Watch For

Preview each LEVER step. Explain that scientists called geneticists study how traits are passed down.

Slide 5: Activity 1: Sort the Parts

7 min

Say: *"What parts of this system can we identify? Where do the traits come from?"*

Watch For

Guide sorting of three components. Explain parent traits are external because they exist before the offspring is born.

Slide 6: Activity 2: Connect with Arrows

7 min

Say: *"When parents have very different traits, does variation go up or down? What about offspring appearance?"*

Watch For

Help students draw arrows showing parent traits drive genetic variation, which determines offspring appearance.

Slide 7: Activity 3: Run the Simulation!

8 min

Say: *"Let us test our model! What happens when similar parents have babies versus very different parents?"*

Watch For

Have students predict first, then run each scenario. Compare offspring from similar vs. different parents.

Slide 8: What Did We Discover?

7 min

Say: *"Now we know — every living thing is unique because of the special MIX of traits from both parents!"*

Watch For

Summarize discoveries. Do a class trait survey (hair color, eye color, attached/detached earlobes) and chart the variation.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Parent Traits
- **INTERNAL:** Genetic Variation, Offspring Appearance

Relationships

- **Parent Traits to Genetic Variation:** POSITIVE (+) — When parents have more diverse traits between them, there are more possible combinations for offspring. Two very different parents can produce offspring with a wider range of appearances than two very similar parents.
- **Genetic Variation to Offspring Appearance:** POSITIVE (+) — More genetic variation means more possible outcomes for what offspring look like. High variation means each sibling is likely to look noticeably different, while low variation means siblings will look more similar to each other.

Key Discoveries

- Offspring inherit traits from BOTH parents, which is why children look similar to their parents but not identical
- Genetic variation means each offspring gets a unique combination of traits — that is why siblings look different
- Some traits are inherited (eye color, hair type) while others are influenced by the environment (scars, learned skills)
- Variation in a group is normal and healthy — it helps species survive changes in their environment

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Children are exact copies of one parent"	Children inherit traits from BOTH parents, not just one. The combination is unique each time, which is why siblings look different. You might have your mom's eyes and your dad's smile — each trait can come from either parent.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"All differences between individuals come from their genes"	While many traits are inherited, some differences come from the environment. A plant grown in sunlight may be shorter and greener than the same plant grown in shade. Scars, learned skills, and muscle development are all environmental, not inherited.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Siblings should look the same since they have the same parents"	Each sibling gets a DIFFERENT random combination of parent genes. With thousands of genes mixing in different ways each time, the chances of two siblings being identical are incredibly small. It is like shuffling a deck of cards — you almost never get the same hand twice.